A Physical attribution: implemented core

Solid arrows: current information flow

PMU reports



8 PMUs; 30 Hz

$|V|, \angle V, |I|, \angle I$

$f, \dot{f}$; 47 coordinates

Mapped terminals

Causal instrument memory

Prefix: fit context

Reports: update evidence

Network predictors



$\dot{x} = f_q(x, z, u)$

$0 = g_q(x, z, u)$

$I_{\text{term}} = Y_{\text{term},q} V$

Local switched operators

Fixed network alternatives

Conditional histories

No new event / active / recovery

$\xi \mid a, h \sim \mathcal{N}(m_h + d_h a, P_\rho)$

$L_{h,k} = \int p(r_k \mid h, a)\, p_h(a)\, da$

Correct states; weight histories

Replay replaces a suffix

Shared covariance approximation

Retain state through OFF

Physical decisions

M_c: active intervention

Γ_c: originating cause

Threshold selected on CAL

Temporal confirmation

Type and source

Onset / restoration

Emission time

Current runs: no online noise or impedance adaptation.

Next report only

B Proposed extension: not enabled in reported experiments

Propagate uncertainty

$P^-_{h,k} = F_h P^+_{h,k-1} F_h^\top + Q_h$

Preserve state–severity dependence and covariance at history fusion

Verify against small exact models before adapting noise

Adapt noise with safeguards

$R_k = U \Lambda_k U^\top + D_k$

Quiet-history eligibility + quality flags

Bounded, slow covariance update

Score with R_k first; learn R_{k+1}

Freeze Q, physical parameters and center

Revisit a bounded window

$p(h, x_{k-L:k} \mid y_{1:k})$

Saved boundary distribution

Stored covariance versions

Replace evidence; never add twice

Report revised onset at current time

C Cyber analysis: separate layer, outside this physical study

Network telemetry + report metadata

External integrity / cyber methods

Interface: timestamp + quality / availability

Any mask requires a declared likelihood

Separate cyber alerts and validation

No cyber score added to event evidence

Solid, dashed and gray regions distinguish implementation, research proposals and an external service boundary.